SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	C. Tharun	**Reg. No.:**	192472309
Section / Batch:	Slot B	**Date:**	13-07-2026

Code of Conduct

I C. Tharun/192472309 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: C. Tharun

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.
Identify email addresses.
Extract mobile numbers.
Detect technical skills (Python, Java, SQL, Machine Learning, NLP).
Extract years of experience.
Generate a structured summary of the candidate's profile.
Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

PROCEDURE:

- Import the re (Regular Expression) module.
- Read the register number, email, course code, semester, and mobile number from the user.
- Validate each input using appropriate regular expressions.
- Display whether each field is valid or invalid.
- If all fields are valid, display "Registration Successful"; otherwise, display "Registration Failed".

CODE:

```
import re
name = input("Enter Name: ")
email = input("Enter Email: ")
mobile = input("Enter Mobile Number: ")
skills = input("Enter Skills: ")
exp = int(input("Enter Experience (years): "))

found_skills = re.findall(r"Python|Java|SQL|Machine Learning|NLP", skills, re.I)

print("\n--- Candidate Summary ---")
print("Name:", name)
print("Email:", email)
print("Mobile:", mobile)
print("Skills:", found_skills)
print("Experience:", exp, "years")
if exp >= 2 and any(skill.lower() == "python" for skill in found_skills):
    print("Eligible")
else:
    print("Not Eligible")
```

Input:

Enter Name: Tharun
Enter Email: tharun54@gmail.com
Enter Mobile Number: 9876543210
Enter Skills: Python Java SQL
Enter Experience (years): 3

Output:

Name: Tharun

Email: tharun54@gmail.com
Mobile: 9876543210
Skills: ['Python', 'Java', 'SQL']
Experience: 3 years
Eligible

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

PROCEDURE:

- Import the re (Regular Expression) module.
- Read the product names and search keyword from the user.
- Select the required search type.
- Use Regular Expressions to search for matching products.
- Display the matching product names.
- Display the total number of matching products.

CODE:

```
import re
products = input("Enter products (comma separated): ").split(",")
key = input("Enter keyword: ")

print("1.Exact  2.Prefix  3.Suffix  4.Partial  5.Case-Insensitive")
ch = int(input("Enter choice: "))
result = []
for p in products:
    p = p.strip()
    if ch == 1 and re.fullmatch(key, p):
        result.append(p)
    elif ch == 2 and re.match(key, p):
        result.append(p)
    elif ch == 3 and re.search(key + "$", p):
        result.append(p)
    elif ch == 4 and re.search(key, p):
        result.append(p)
    elif ch == 5 and re.search(key, p, re.I):
        result.append(p)
print("Matching Products:", result)
print("Total Matches:", len(result))
```

Input:
Enter products (comma separated): Laptop, Phone, Laptop Bag, Smart Phone, Mouse
Enter keyword: Laptop

1.Exact 2.Prefix 3.Suffix 4.Partial 5.Case-Insensitive
Enter choice: 2

Output:
Matching Products: ['Laptop', 'Laptop Bag']
Total Matches: 2

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information.
Validate the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

PROCEDURE:

- Import the re (Regular Expression) module.
- Read the register number, institutional email, course code, semester, and mobile number from the user.
- Validate each field using appropriate Regular Expressions.
- Display whether each field is valid or invalid.
- Display the final registration status.

CODE:

```
import re
reg = input("Register No: ")
email = input("Email: ")
course = input("Course Code: ")
sem = input("Semester: ")
mobile = input("Mobile: ")
print("Register:", "Valid" if re.fullmatch(r"\d{12}", reg) else "Invalid")
print("Email:", "Valid" if re.fullmatch(r"\w+@\w+\.edu", email) else "Invalid")
print("Course:", "Valid" if re.fullmatch(r"[A-Z]{3}\d{3}", course) else "Invalid")
print("Semester:", "Valid" if re.fullmatch(r"[1-8]", sem) else "Invalid")
print("Mobile:", "Valid" if re.fullmatch(r"[6-9]\d{9}", mobile) else "Invalid")

if (re.fullmatch(r"\d{12}", reg) and
    re.fullmatch(r"\w+@\w+\.edu", email) and
    re.fullmatch(r"[A-Z]{3}\d{3}", course) and
    re.fullmatch(r"[1-8]", sem) and
    re.fullmatch(r"[6-9]\d{9}", mobile)):
    print("Registration Successful")
else:
    print("Registration Failed")
```

Input:
Register No: 123456789012
Email: 123456789012@college.edu
Course Code: CSE101

Semester: 5
Mobile: 9876543210

Output:
Register: Valid
Email: Valid
Course: Valid
Semester: Valid
Mobile: Valid
Registration Successful

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________